

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) Science — Semester I Examination, 2023-24

**Paper No. MTB-101: Calculus-I**

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 067931

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

**Question 1.** (Compulsory — 2 marks each.)

- (a) Find the least  $m \in \mathbb{N}$  for the sequence  $x_n = \frac{3n+1}{7n-4}$  such that  $\left| \frac{3n+1}{7n-4} - \frac{3}{7} \right| < \varepsilon$  for all  $n > m$ .
- (b) Check whether  $\{x_n\}$  defined by  $x_1 = \sqrt{7}$ ,  $x_{n+1} = \sqrt{7+x_n}$  is monotonically increasing or decreasing.
- (c) Examine convergence of  $\sum_{n=1}^{\infty} \frac{n}{n(n+1)(n+3)}$ .
- (d) If  $\sum u_n$  converges with positive terms, prove  $\sum u_{2n}$  converges.
- (e) Let  $f : I \rightarrow \mathbb{R}$ ,  $\lim_{x \rightarrow c} f(x) = \ell$ . Show  $\lim_{x \rightarrow c} |f(x)| = |\ell|$ .
- (f) Give an example of  $f, g$  not continuous at  $c$  but  $fg$  continuous at  $c$ .
- (g)  $f : \mathbb{R} \rightarrow \mathbb{R}$  satisfies  $f(x+y) = f(x)f(y)$ . If  $f$  is differentiable at 0, prove  $f'(c) = f(c)f'(0)$ .
- (h) Let  $f : I \rightarrow \mathbb{R}$  be differentiable at  $c$  with  $f'(c) < 0$ . Show  $f$  is decreasing at  $c$ .
- (i) If  $c_0 + \frac{c_1}{2} + \frac{c_2}{3} + \cdots + \frac{c_n}{n+1} = 0$ , show  $c_0 + c_1x + \cdots + c_nx^n = 0$  has a root in  $(0, 1)$ .
- (j) Let  $y = e^x(\cos x + \sin x)$ . Prove  $y'' + 4y = 0$ .
- (k) Draw the graph of  $r = 2 \sin(2\theta)$ .

**Question 2.**

- (a) Let  $\{x_n\}, \{y_n\}$  converge to  $x, y$  (non-zero). Show  $\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \frac{x}{y}$  provided  $y \neq 0$ . [6]
- (b) Let  $x_{n+1} = \frac{1}{2}(x_n + y_n)$ ,  $y_{n+1} = \sqrt{x_n y_n}$  for  $n \geq 1$ ,  $x_1, y_1 > 0$ . Prove  $\lim x_n = \lim y_n$ . [6]

**Question 3.**

- (a) Let  $\sum u_n, \sum v_n$  be positive-term series with  $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \ell \neq 0$ . Prove they converge or diverge together. [6]
- (b) Test convergence of  $\sum_{n=0}^{\infty} \frac{n^2 - 1}{n^2 \cdot 2^n}$ . [3]
- (c) Give  $\varepsilon$ - $\delta$  definition of  $\lim_{x \rightarrow c} f(x) = \ell$ . Give geometrical representation. Show  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ . [3]

**Question 4.**

- (a) Let  $f : I \rightarrow J$  continuous at  $c$ ,  $g : J \rightarrow \mathbb{R}$  continuous at  $f(c)$ . Show  $g \circ f$  is continuous at  $c$ . [6]
- (b) Explain differentiability of  $f$  at  $c$ . If  $f$  differentiable at  $c$ , show  $f$  continuous at  $c$ . Give an example of a function continuous but not differentiable at a point in  $[-4, 4]$ . [6]
- (c) Let  $f : [a, b] \rightarrow \mathbb{R}$  be differentiable with  $f'(a) \neq f'(b)$ . If  $u$  lies between  $f'(a)$  and  $f'(b)$ , prove  $\exists c \in (a, b)$  with  $f'(c) = u$ . [6]

**Question 5.**

- (a) State Rolle's theorem. Verify it for  $f(x) = x \sin \frac{1}{x}$  ( $x \neq 0$ ),  $f(0) = 0$ . [6]
- (b) State and prove Lagrange's Mean Value Theorem. Verify with an example. [6]

**Question 6.**

- (a) Find the Maclaurin expansion of  $y = e^x \sin x$  in powers of  $x$ . [6]
- (b) Find the  $n$ -th derivative of  $f(x) = \log(5x + 3) \cdot \cos(3x + 5)$ . [6]

**Question 7.**

- (a) Find the area under  $y = x^2$  from  $x = \frac{1}{2}$  to  $x = 2$  using limit of a sum. [6]
- (b) Find asymptotes of  $y = \frac{x^2 + 2x - 1}{x}$  and trace the curve. [6]

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